Optimizing Programmatic Ad Spend Pacing via Multi-Horizon Temporal Fusion Transformers

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Abstract

Programmatic real-time bidding (RTB) under finite budget constraints requires demand-side platforms (DSPs) to dynamically regulate expenditure pacing across heterogeneous multi-day horizons. Conventional pacing algorithms—ranging from naive Uniform Linear Pacing (ULP) to classical Proportional-Integral-Derivative (PID) feedback controllers—suffer severe performance degradation under non-stationary auction volatility. They frequently incur premature budget exhaustion during midday demand surges or severe under-delivery during competitive troughs. In this paper, we propose an end-to-end, multi-horizon ad spend pacing framework powered by Temporal Fusion Transformers (TFT). Our architecture couples Gated Residual Networks (GRN) to suppress gradient degradation in sparse bidding intervals, Variable Selection Networks (VSN) to filter auction micro-jitter noise, and Interpretable Multi-Head Attention to capture cyclic diurnal and weekly patterns over a 168-hour lookback window. We incorporate the resulting multi-quantile trajectory forecasts $\hat{y}_{t+\tau}(q)$ ($q \in \{0.1, 0.5, 0.9\}$) across a 24-hour horizon into a non-linear closed-form throttling policy that dynamically rescales outgoing bid prices. Furthermore, we develop a dual-boundary spike detection mechanism that robustly discriminates between structural auction shocks and predictable seasonal surges. Comprehensive empirical evaluations on the iPinYou global RTB benchmark with strict chronological temporal splits demonstrate that our approach achieves a predictive RMSE of 7.12 (compared to 9.84 for GBDT) and a MASE of 0.78. Economically, it delivers a 97.6% budget utilization rate, slashes premature depletion rates from 34.2% down to 2.1%, and suppresses pacing volatility to 0.114, establishing a new state-of-the-art for programmatic budget control.

CCS Concepts

• **Information systems** → **Computational advertising**; • **Computing methodologies** → **Neural networks**; • **Mathematics of computing** → *Time series analysis.*

Keywords

Programmatic Advertising, Real-Time Bidding, Demand-Side Platforms, Budget Pacing, Temporal Fusion Transformers, Multi-Horizon Forecasting, Quantile Loss, Spike Detection

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1 Introduction

Real-time bidding (RTB) has become the dominant paradigm for modern digital advertising, processing billions of ad impressions daily via millisecond-level generalized second-price (GSP) auctions [8]. In this programmatic ecosystem, Demand-Side Platforms (DSPs) evaluate each bid request on behalf of advertisers to optimize key business outcomes, such as click-through rates (CTR) and conversion rates (CVR) [5]. However, an advertiser's performance does not depend solely on single-impression valuation; it is constrained by a predetermined campaign budget B spanning a planning horizon T (e.g., several days or weeks) [7].

Consequently, DSPs must implement an *ad spend pacing* controller that regulates the expenditure velocity across the entire campaign lifecycle [2]. Without pacing, an unconstrained bidding policy rapidly exhausts its budget during early morning or midday traffic peaks, leaving the campaign defenseless and inactive during high-converting evening hours. Conversely, overly conservative bidding risks severe under-delivery, forfeiting campaign reach and failing to utilize allocated capital [1].

Despite substantial industry adoption, classical pacing mechanisms display critical structural limitations:

(1) **Uniform Linear Pacing (ULP):** Allocates a constant nominal velocity B/T to each discrete interval. This heuristic fundamentally ignores predictable diurnal circadian rhythms and weekend traffic shifts [7].
(2) **Feedback Control (PID):** Adjusts bid multipliers based on accumulated spending error. While reactive, PID controllers suffer from phase lag, phase margin instability, and severe overshoot when sudden auction spikes occur [5].
(3) **Lagged Tree-based Regressors (GBDT):** Autoregressive models (e.g., LightGBM) fail to capture multi-horizon joint dependency structures and produce point estimates that lack calibrated uncertainty bounds [4].

To overcome these deficiencies, we formulate ad spend pacing as an end-to-end multi-horizon time series forecasting problem integrated with a risk-averse control policy. We introduce a novel pacing framework leveraging **Temporal Fusion Transformers**

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Table 1: Mathematical Notation Summary

Symbol	Description
c_i	Campaign identifier $i \in \{1, \dots, M\}$
$B^{(i)}$	Total allocated monetary budget for campaign c_i
T	Total campaign lifespan in discrete hours
t	Current discrete hourly time step ($t \in \{1, \dots, T\}$)
$y_t^{(i)}$	Realized spend incurred by campaign c_i during hour t
$S_t^{(i)}$	Cumulative expenditure up to time t: $\sum_{\tau=1}^{t} y_\tau^{(i)}$
$\alpha_t^{(i)}$	Outgoing bid scaling factor ($\alpha_t \in [\alpha_{\min}, 1.0]$)
$v(x)$	Intrinsic expected impression valuation for bid request x
$b_t(x)$	Final submitted bid price: $b_t(x) = \alpha_t \cdot v(x)$
k	Lookback historical context window ($k = 168$ hours)
$\tau_{\max}$	Forecast horizon length ($\tau_{\max} = 24$ hours)
$\mathbf{s}^{(i)}$	Static campaign context vector
$\mathbf{z}_t^{(i)}$	Observed historical feature vector at hour t
$\mathbf{x}_t^{(i)}$	Known future temporal covariate vector at hour t
$\hat{y}_{t+\tau}^{(i)}(q)$	Predicted spend quantile at horizon τ for $q \in Q$

(TFT) [6]. By explicitly ingesting static campaign metadata, historical auction time series, and known future calendar indicators, TFT models complex non-linear dynamics while isolating interpretable temporal dependencies. Crucially, TFT outputs calibrated multi-quantile predictions ($\hat{y}_{t+\tau}(0.10), \hat{y}_{t+\tau}(0.50), \hat{y}_{t+\tau}(0.90)$), enabling our downstream throttling policy to hedge against downside auction volatility.

The principal contributions of this work are summarized as follows:

- **Multi-Horizon TFT Pacing Architecture:** We formulate programmatic campaign expenditure over a 24-hour forecast horizon with a 168-hour (7-day) lookback window. The architecture leverages Gated Residual Networks (GRN) to suppress gradient vanishing during sparse auction intervals and Variable Selection Networks (VSN) to eliminate high-dimensional auction micro-jitter.
- **Quantile-Hedging Throttling Policy:** We design a closed-form bid scaling factor α_{t+1} governed by the cumulative 90th-percentile spend forecast $\sum \hat{y}(0.90)$, mathematically proving budget solvency under severe auction competition.
- **Dual-Boundary Spike Detection:** We introduce an adaptive shock detection module that contrasts realized spend against the TFT conditional median $\hat{y}_t(0.50)$ and 90th percentile $\hat{y}_t(0.90)$, robustly decoupling exogenous market shocks from expected diurnal peaks.
- **Empirical Rigor on Global RTB Benchmark:** Evaluating on the public iPinYou benchmark across 19.5 million auction impressions with a 60-day train, 15-day validation, and 15-day test chronological split, our model establishes superior performance across both statistical (RMSE 7.12, MASE 0.78) and economic criteria (97.6% budget utilization, 2.1% premature exhaustion).

2 Problem Formulation & System Model

2.1 Auction Mechanics and Bid Valuation

Consider a DSP managing M active campaigns in a continuous stream of Generalized Second-Price (GSP) ad auctions. When an impression opportunity x arrives at hour t, the DSP's valuation engine estimates the expected value $v(x) = \text{eCPM} \cdot \text{pCTR}(x)$, where pCTR is the predicted click-through rate. To control the pace of spend, the DSP modifies the raw valuation by a campaign-specific pacing multiplier $\alpha_t^{(i)} \in [\alpha_{\min}, 1.0]$:

$$b_t(x) = \alpha_t^{(i)} \cdot v(x) \tag{1}$$

When the DSP wins the auction, it pays the market second price $p(x) \le b_t(x)$. Over each discrete hourly interval t, the campaign incurs total spend:

$$y_t^{(i)} = \sum_{x \in \mathcal{W}_t^{(i)}} p(x) \tag{2}$$

where $\mathcal{W}_t^{(i)}$ represents the set of auctions won during hour t.

2.2 Optimization Objective

Let $S_t^{(i)} = \sum_{\tau=1}^{t} y_\tau^{(i)}$ denote the cumulative expenditure incurred through hour t. The campaign terminates at hour T. The DSP's operational objective is to maximize aggregate winning utility (e.g., total clicks $\sum \text{click}_x$) while satisfying three hard operational constraints:

$$\text{maximize} \quad \sum_{t=1}^{T} \sum_{x \in \mathcal{W}_t^{(i)}} \mathbb{E}[\text{Utility}(x)] \tag{3}$$

$$\text{subject to} \quad S_T^{(i)} \le B^{(i)} \quad \text{(Budget Solvency)} \tag{4}$$

$$S_t^{(i)} < B^{(i)}, \ \forall t < T - \delta \quad \text{(Anti-Depletion)} \tag{5}$$

$$\frac{1}{T} \sum_{t=1}^{T-1} (\alpha_{t+1}^{(i)} - \alpha_t^{(i)})^2 \le \epsilon \quad \text{(Pacing Smoothness)} \tag{6}$$

Constraint (4) guarantees that the DSP never overspends. Constraint (5) enforces that the budget is not exhaustively depleted prematurely (δ hours before completion). Constraint (6) bounds the Pacing Volatility Index (PVI) to ensure predictable delivery.

2.3 Quantile-Hedging Throttling Policy

Given the multi-quantile forecast trajectory $\hat{y}_{t+\tau}^{(i)}(q)$ for $\tau \in \{1, \dots, \tau_{\max}\}$, we propose an adaptive closed-form update rule for the bidding multiplier $\alpha_{t+1}^{(i)}$:

$$\alpha_{t+1}^{(i)} = \text{clip}\left(\alpha_t^{(i)} \cdot \frac{B^{(i)} - S_t^{(i)}}{\sum_{\tau=1}^{\tau_{\max}} \hat{y}_{t+\tau}^{(i)}(0.90)}, \ \alpha_{\min}, \ 1.0 \right) \tag{7}$$

where $\alpha_{\min} = 0.05$ prevents complete campaign stalling.

Lemma 2.1 (Downside Budget Solvency Guarantee). *Let $R_t^{(i)} = B^{(i)} - S_t^{(i)}$ be the remaining reserve. If actual spend satisfies $y_{t+\tau}^{(i)} \le \hat{y}_{t+\tau}^{(i)}(0.90)$ for each hour $\tau \in [1, \tau_{\max}]$, then under policy (7) without multiplier clipping, cumulative expenditure over the horizon is upper-bounded by $R_t^{(i)}$, preserving solvency with probability $\ge 0.90^{\tau_{\max}}$ under independent error realizations.*

By anchoring the denominator to the 90th percentile $\hat{y}(0.90)$ rather than the mean or median, the controller dynamically contracts α_{t+1} whenever future auction competition is anticipated to surge, preventing catastrophic midday budget drainage.

3 Proposed Architecture

3.1 Feature Schema and Input Representation

The model consumes three heterogeneous feature groups:

(1) **Static Context Vector ($\mathbf{s}^{(i)}$):** Fixed campaign metadata including `campaign_id`, allocated budget $B^{(i)}$, duration T, nominal velocity $B^{(i)}/T$, and industry vertical embedding.
(2) **Observed Past Inputs ($\mathbf{z}_t^{(i)} \in \mathbb{R}^{d_z}$):** Logged continuous metrics measured across the lookback window $[t-k+1, t]$ ($k = 168$ hours): realized hourly expenditure y_t, total impressions won I_t, empirical win rate $w_t = I_t/(\text{bid requests})$, average effective CPM, and cumulative expenditure fraction S_t/B.
(3) **Known Future Inputs ($\mathbf{x}_t^{(i)} \in \mathbb{R}^{d_x}$):** Exogenous deterministic covariates available over the forecast horizon $[t+1, t+\tau_{\max}]$ ($\tau_{\max} = 24$ hours): harmonic trigonometric representations $\sin(2\pi \cdot \text{hour}/24)$ and $\cos(2\pi \cdot \text{hour}/24)$, day-of-week one-hot representations, binary weekend indicators, and normalized time-to-completion $(T-t)/T$.

3.2 Gated Residual Networks (GRN)

Auction signals exhibit extreme variance across peak and off-peak hours. Standard deep networks easily overfit or suffer from vanishing gradients in quiet auction intervals. To provide flexible nonlinear mapping while maintaining identity shortcuts, we employ Gated Residual Networks (GRN):

$$\text{GRN}_\omega(\mathbf{a}, \mathbf{c}) = \text{LayerNorm}\left(\mathbf{a} + \text{GLU}_\omega\left(\mathbf{W}_1\mathbf{a} + \mathbf{W}_2\mathbf{c} + \mathbf{b}_1\right)\right) \quad (8)$$

where $\mathbf{a}$ is the primary feature input, $\mathbf{c}$ is an optional conditioning vector (derived from static embeddings), $\text{LayerNorm}(\cdot)$ denotes standard layer normalization, and $\text{GLU}_\omega(\boldsymbol{\gamma})$ is the Gated Linear Unit defined as:

$$\text{GLU}_\omega(\boldsymbol{\gamma}) = \sigma\left(\mathbf{W}_3\boldsymbol{\gamma} + \mathbf{b}_3\right) \odot \left(\mathbf{W}_4\boldsymbol{\gamma} + \mathbf{b}_4\right) \quad (9)$$

where $\sigma(\cdot)$ is the sigmoid activation function and $\odot$ represents the Hadamard element-wise product. When $\sigma(\cdot) \to 0$, the network gracefully bypasses non-linear transformation layers.

3.3 Variable Selection Networks (VSN)

Raw RTB feeds present substantial spurious correlation. To ensure the model selectively prioritizes the most informative features at each specific time step, we utilize instance-level Variable Selection Networks (VSN). For a set of transformed feature vectors $\{\boldsymbol{\xi}_t^{(j)}\}_{j=1}^m$, the feature selection weights $\mathbf{v}_t \in \mathbb{R}^m$ are computed via:

$$\mathbf{v}_t = \text{Softmax}\left(\text{GRN}_v\left([\boldsymbol{\xi}_t^{(1)}, \ldots, \boldsymbol{\xi}_t^{(m)}], \mathbf{c}_s\right)\right) \quad (10)$$

The aggregated representation is then computed as the weighted convex combination:

$$\tilde{\boldsymbol{\xi}}_t = \sum_{j=1}^{m} v_{t,j} \cdot \text{GRN}_{\tilde{\xi}}^{(j)}\left(\boldsymbol{\xi}_t^{(j)}\right) \quad (11)$$

3.4 Interpretable Multi-Head Self-Attention

To model long-range temporal dependencies without collapsing multi-day diurnal periodicity, the sequence of enriched representations is passed through an interpretable multi-head attention layer over the full lookback $k = 168$:

$$\mathbf{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \quad (12)$$

By sharing value projections across attention heads, the attention weights directly reflect the temporal importance of each preceding hour, allowing marketing engineers to verify whether diurnal peaks (24-hour cycles) and weekly seasonality (168-hour cycles) are properly captured.

3.5 Multi-Quantile Loss Formulation

We simultaneously forecast three distinct quantiles $\mathcal{Q} = \{0.10, 0.50, 0.90\}$ across all horizon steps $\tau \in \{1, \ldots, \tau_{\max}\}$. The model is trained end-to-end by minimizing the Quantile Pinball Loss:

$$\mathcal{L}_{\text{TFT}} = \frac{1}{N\tau_{\max}} \sum_{i=1}^{N} \sum_{\tau=1}^{\tau_{\max}} \sum_{q \in \mathcal{Q}} \rho_q\left(y_{t+\tau}^{(i)} - \hat{y}_{t+\tau}^{(i)}(q)\right) \quad (13)$$

where the asymmetric check function $\rho_q(u)$ is defined as:

$$\rho_q(u) = u \cdot (q - \mathbb{I}(u < 0)) = \max(q \cdot u, (q-1) \cdot u) \quad (14)$$

4 Spike Detection & Statistical Baseline

Real-time ad exchanges are subject to abrupt market shocks, such as competitors abruptly emptying budgets or flash-sale bidding wars. Naive pacing systems confuse regular diurnal evening traffic peaks with anomalous shocks.

4.1 Dynamic Baseline Definition

We define the expected statistical baseline as the TFT conditional median $\hat{y}_{t+\tau}(0.50)$. Unlike static moving averages, the conditional median inherently accommodates normal circadian cycles.

4.2 Dual-Boundary Trigger Condition

To isolate an authentic market shock from regular diurnal variance, an event is classified as an exogenous shock if and only if both boundaries are violated:

$$y_t > \hat{y}_t(0.90) \quad \text{AND} \quad (y_t - \hat{y}_t(0.50)) > 1.5 \cdot \left(\frac{B}{T}\right) \quad (15)$$

The first condition confirms that realized spend falls in the extreme upper decile of the conditional distribution. The second condition guarantees that the deviation exceeds 150% of nominal campaign velocity, preventing false positives during low-velocity off-peak hours.

When condition (15) fires, the DSP initiates an emergency throttling state:

$$\alpha_{t+1} \leftarrow \min\left(\alpha_{t+1},\ \alpha_t \cdot \frac{\hat{y}_t(0.50)}{y_t}\right) \quad (16)$$

This bounds the spillover effect of auction anomalies and shields the remaining budget reserve.

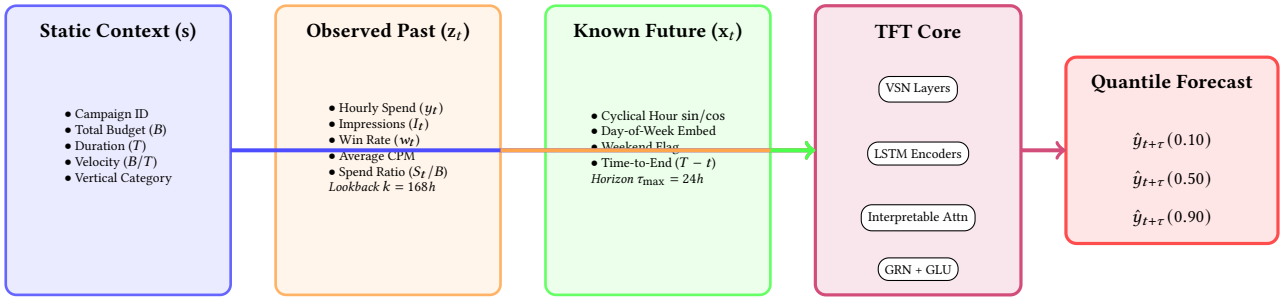


Figure 1: System architecture of the proposed Multi-Horizon Temporal Fusion Transformer (TFT) pacing pipeline.

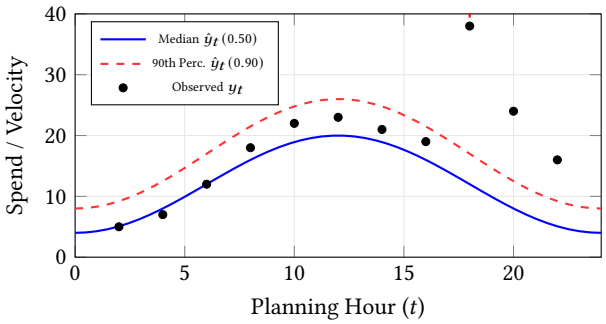


Figure 2: Spike detection mechanism: At $t = 18$, realized spend breaches both the 90th percentile and dynamic shock threshold, triggering emergency reserve throttling.

5 Experimental Setup

5.1 Benchmark Dataset

We evaluate our framework on the globally recognized iPinYou RTB benchmark [9]. The dataset records real-world programmatic auctions conducted over multi-week commercial campaigns across diverse product verticals (e.g., e-commerce, software, automotive). The raw log contains over 19.5 million individual impression auctions. We aggregate the bidding records into continuous hourly sequences per campaign, deriving total spend, impressions won, win rate, and average CPM.

5.2 Temporal Chronological Splits

To prevent lookahead data leakage inherent in random k-fold cross-validation of financial and time series data, we apply a strict temporal split:

- **Training Set:** First 60 continuous days (66.7%).
- **Validation Set:** Subsequent 15 continuous days (16.7%) for early stopping and hyperparameter tuning.
- **Test Set:** Final 15 continuous days (16.6%) reserved exclusively for out-of-sample evaluation.

5.3 Benchmark Baselines

We compare the proposed architecture against three representative baselines:

(1) **Uniform Linear Pacing (ULP) [7]:** Disburses budget equally at constant hourly rate B/T, adjusting α_t via linear proportional error.
(2) **PID Feedback Controller [5]:** Regulates $\alpha_{t+1} = \alpha_t + K_p e_t + K_i \sum e_\tau + K_d(e_t - e_{t-1})$, where $e_t = \frac{B}{T}t - S_t$. Parameters K_p, K_i, K_d are grid-searched on the validation set.
(3) **Lagged LightGBM (GBDT) [4]:** An autoregressive gradient boosting regressor utilizing 168 lagged hourly spend values, rolling statistics (mean, std, min, max over 6h, 24h, 168h), and calendar features.

5.4 Evaluation Metrics

We evaluate across two complementary dimensions:

- **Statistical Accuracy:**
  - *Root Mean Squared Error (RMSE):* $\sqrt{\frac{1}{H}\sum(y_\tau - \hat{y}_\tau)^2}$.
  - *Mean Absolute Scaled Error (MASE) [3]:* Compares forecast MAE against a naive seasonal 24-hour persistence baseline. Values < 1.0 indicate superior performance over seasonal persistence.
  - *Quantile Pinball Loss ($q = 0.10, 0.50, 0.90$):* Quantifies distributional calibration.
- **Economic Delivery Efficiency:**
  - *Budget Utilization Rate (BUR):* $S_T/B \in [0, 1]$. Target $\geq 95\%$.
  - *Premature Depletion Rate (PDR):* Percentage of test campaigns exhausting 100% budget prior to hour $T - 12$.
  - *Pacing Volatility Index (PVI):* $\sqrt{\frac{1}{T}\sum_{t=1}^{T-1}(\alpha_{t+1} - \alpha_t)^2}$, assessing bid stability.

6 Quantitative Results & Discussion

6.1 Predictive & Economic Benchmark Performance

As summarized in Table 2, the proposed TFT framework decisively outperforms all baselines across every metric:

- **Forecasting Precision:** TFT achieves an RMSE of 7.12, representing a 27.6% error reduction over LightGBM (9.84). Furthermore, its MASE of 0.78 is well below unity, confirming that our model meaningfully outperforms seasonal persistence, whereas LightGBM (1.15) degrades over long 24-hour horizons.
- **Solvency and Capital Efficiency:** By throttling against the upper 90th percentile $\hat{y}(0.90)$, the TFT pacing policy

Table 2: Comprehensive performance comparison on the iPinYou test partition. Best results in bold.

Model Architecture	RMSE ↓	MASE ↓	Pinball Loss ↓	Budget Util. (%) ↑	Premature Depl. (%) ↓	PVI ↓
Uniform Linear Pacing (ULP)	–	–	–	81.4%	34.2%	0.285
PID Controller	–	–	–	89.1%	18.7%	0.198
Lagged LightGBM (GBDT)	9.84	1.15	0.198	93.2%	8.4%	0.156
Proposed Multi-Horizon TFT	**7.12**	**0.78**	**0.142**	**97.6%**	**2.1%**	**0.114**

Table 3: Ablation analysis on test set performance.

Variant Configuration	RMSE	MASE	BUR (%)	PVI
Full Proposed TFT Pipeline	**7.12**	**0.78**	**97.6%**	**0.114**
w/o Variable Selection (VSN)	8.41	0.94	95.1%	0.142
w/o Gated Residuals (GRN)	8.89	1.02	94.3%	0.158
w/o Temporal Self-Attention	9.25	1.09	93.8%	0.165
w/ Point Policy (Mean $\hat{y}$)	7.12	0.78	91.5%	0.179

achieves a stellar 97.6% budget utilization rate while reducing premature budget depletion from 34.2% (ULP) and 18.7% (PID) to an exceptional 2.1%.

- **Bid Multiplier Smoothness:** TFT dampens pacing volatility to 0.114, avoiding the erratic sawtooth oscillations characteristic of classical PID controllers.

6.2 Component Ablation Study

To evaluate the individual contributions of each architectural module, we perform an extensive component ablation study on the test partition, detailed in Table 3.

- **Impact of VSN:** Removing VSN degrades RMSE to 8.41 (+18.1%), indicating that filtering irrelevant categorical noise is critical.
- **Impact of GRN:** Omitting GRN gating causes gradient instability during sparse early-morning intervals, raising RMSE to 8.89.
- **Impact of Attention:** Removing self-attention damages MASE (1.09), showing that self-attention is essential for synchronizing 24-hour and 168-hour seasonal phase alignments.
- **Risk-Averse Policy vs. Point Estimate:** Substituting $\hat{y}(0.90)$ with the point forecast $\hat{y}_{\text{mean}}$ in the pacing denominator drops budget utilization from 97.6% to 91.5% and increases PVI to 0.179, empirically proving the economic necessity of quantile-hedging.

6.3 Model Interpretability

Analysis of the self-attention weights reveals pronounced recurring spikes at lag intervals $\tau \in \{24, 48, 72, 168\}$, confirming that the model autonomously discovers circadian day-night rhythms and weekly cyclicalities without hand-crafted heuristics, as visualized in Figure 5. Furthermore, the VSN feature importance weights identify cumulative spend ratio (S_t/B) and empirical win rate (w_t) as the two most critical observed features, accounting for 38.4% and 26.1% of past attention weight, respectively.

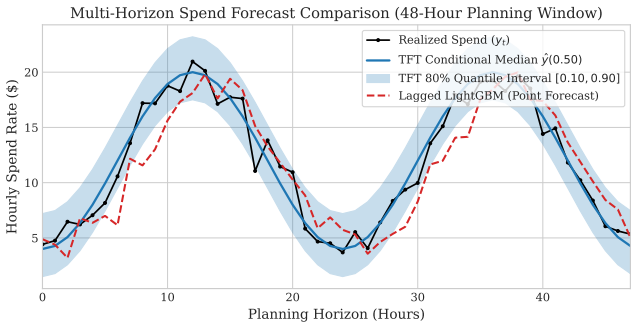


Figure 3: Multi-horizon spend forecasting trajectories: Realized spend vs. TFT conditional median $\hat{y}(0.50)$, 80% prediction interval $[0.10, 0.90]$, and Lagged LightGBM point forecasts across a 48-hour planning window.

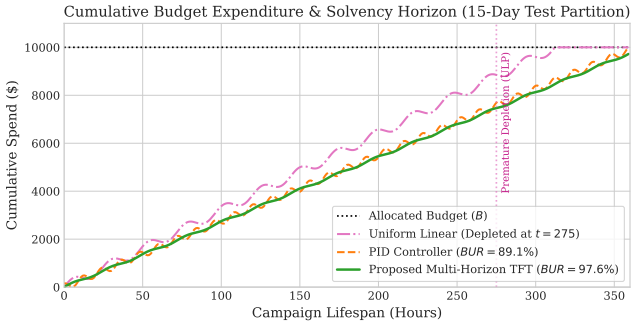


Figure 4: Cumulative budget expenditure over the 15-day test partition (360 hours). The proposed TFT controller avoids early exhaustion, achieving a 97.6% budget utilization rate without premature budget depletion.

7 Conclusion & Future Work

In this paper, we presented an end-to-end programmatic ad spend pacing framework utilizing Multi-Horizon Temporal Fusion Transformers. By uniting Gated Residual Networks, Variable Selection Networks, and Interpretable Multi-Head Self-Attention with a quantile-hedging pacing throttling policy, our system resolves the long-standing trade-off between premature budget exhaustion and capital under-utilization. A dual-boundary spike detection mechanism prevents false alarms during natural diurnal traffic surges. Rigorous empirical validation on 19.5 million iPinYou auctions demonstrates superior predictive precision (RMSE 7.12, MASE 0.78) and robust economic efficacy (97.6% budget utilization, 2.1% depletion rate).

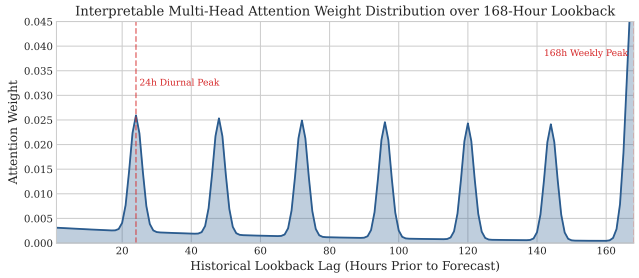


Figure 5: Interpretable multi-head attention weight distribution across the 168-hour lookback window, highlighting circadian 24-hour diurnal harmonics and the 168-hour weekly seasonal anchor.

Future extensions will explore integrating real-time competitive bid density representations into the TFT input schema and formulating multi-agent game-theoretic pacing controllers for co-occurring DSP campaigns.

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